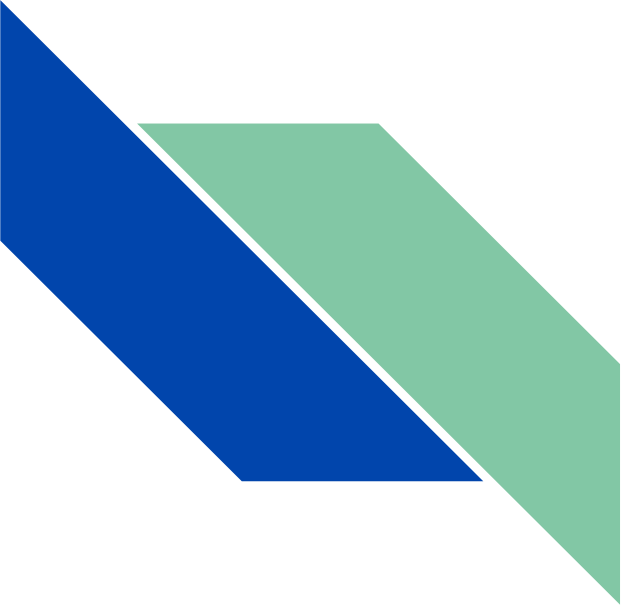




Explainable AI for Post-COVID Psychological Profile Prediction

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Advisor: Assistant Professor Paul Rossener Regonia, Ph.D.



Overview

- Significance of the Study
- Project Objectives
- State of the Art
- Proposed Methodology
- Timeline
- Summary
- References

Significance of the Study



Significance of the Study

Phase 1
May 11-12, 2020

*Participants who lived in
Tokyo, Osaka, Hyogo, and Fukuoka*
n = 6723

Phase 2
June 14-20, 2021

Response
n = 4592

No response
n = 2131

Phase 3
May 13-30, 2022

Response
n = 3892

No response
n = 700

Phase 4
May 10-17, 2024

Response
n = 2659

No response
n = 1233





Significance of the Study

K6 (Kessler et al., 2010): 0-12: Unlikely distress 13-24: Likely distress	PHQ-9 (Kroenke et al., 2001): 0-4: Minimal depression 5-9: Mild 10-14: Moderate 15-19: Moderately severe 20-27: Severe	GAD-7 (Generalised Anxiety Disorder Assessment, n.d.): 0-4: Minimal anxiety 5-9: Mild 10-14: Moderate 15-21: Severe
SSS-8 (Gierk et al., 2014): 0-3: Minimal somatic symptom burden 4-7: Low 8-11: Medium 12-15: High 16-32: Very high	UCLA Loneliness Scale (Deckx et al., 2014): 0-19: Minimal loneliness 20-34: Low 35-49: Moderate 50-64: Moderately high 65-80: High	



Significance of the Study

Null Hypothesis:

There is no significant difference in these mental health metrics during the COVID-19 pandemic and after it.

Alternative Hypothesis (One-Sided Test):

There is a statistically significant increase in these mental health metrics after COVID-19.



Significance of the Study

K6 (Kessler et al., 2010): p-value = 0.00367	PHQ-9 (Kroenke et al., 2001): p-value = 0.00046	GAD-7 (GAD Assessment, n.d.): p-value = 0.00000017
SSS-8 (Gierk et al., 2014): p-value = 0.01268	UCLA Loneliness (Deckx et al., 2014): p-value = 0.00000002	

p-value < 0.05



Significance of the Study

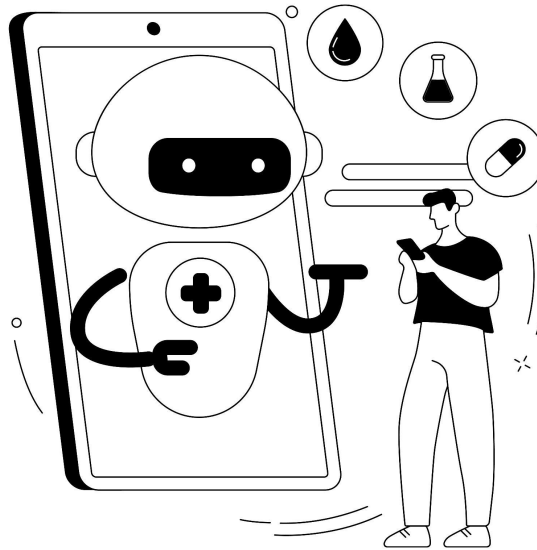
Null Hypothesis:

~~There is no significant difference in these mental health metrics during the COVID-19 pandemic and after it.~~

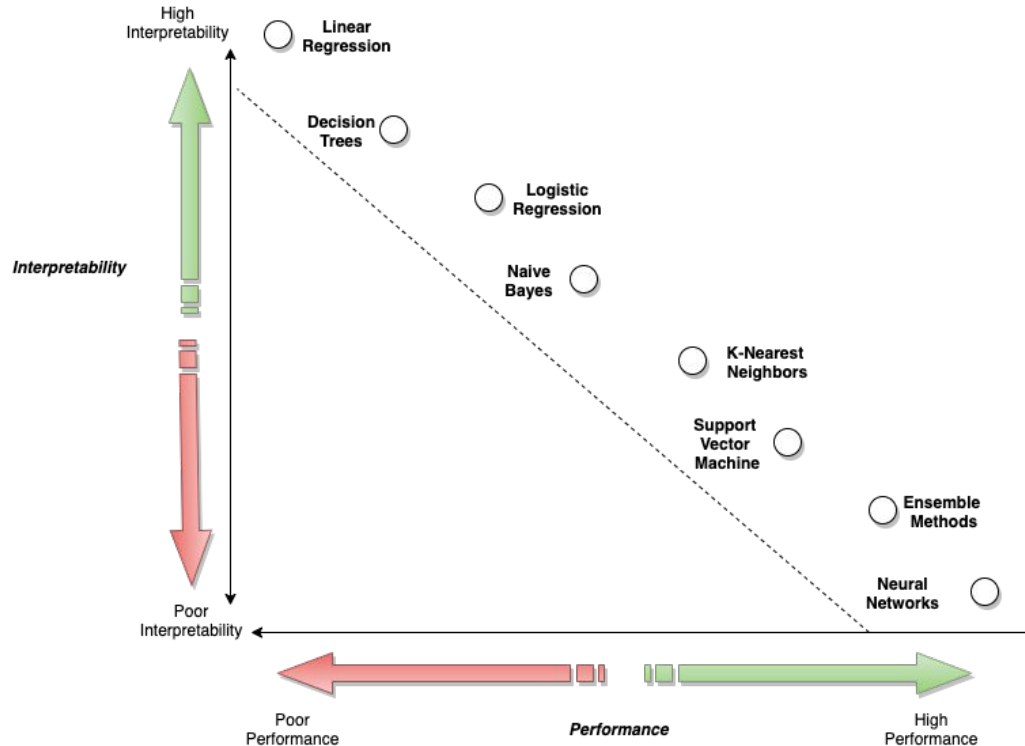
Alternative Hypothesis (One-Sided Test):

There is a statistically significant increase in these mental health metrics after COVID-19.

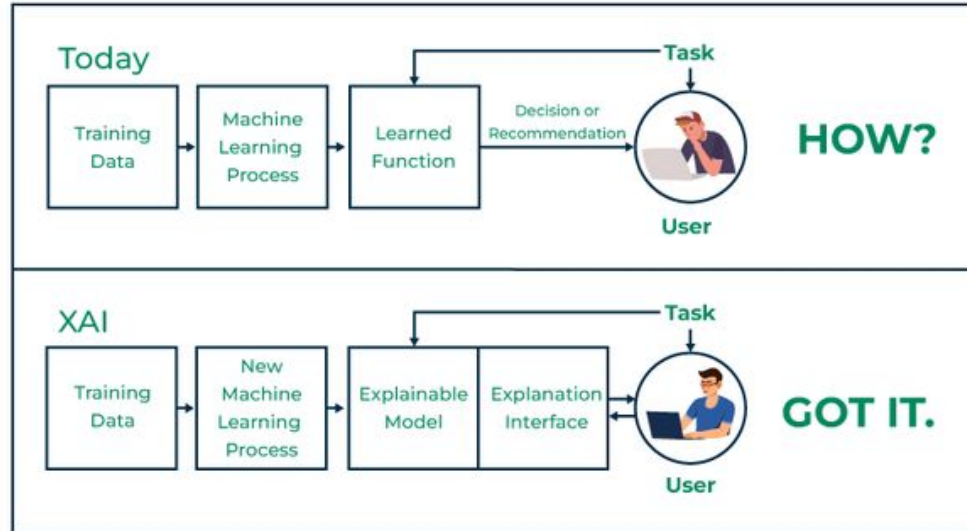
Significance of the Study



Significance of the Study



Significance of the Study





Significance of the Study

“If X had not occurred,
then Y would not have occurred”

“If I hadn’t taken a sip of this hot coffee,
then I wouldn’t have burned my tongue.”

Model

Classification

[illegible]

Black Box

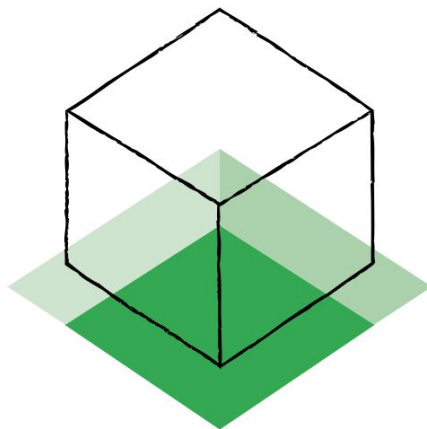
K6: 15.5
PHQ-9: 11.3
GAD-7: 12.6
SSS-8: 12.8
UCLA Loneliness Scale: 37.9

Project Objectives

Data

SAMPLEID	1_ANSWERS1_SEX	1_AGE	1_AGEID	1_PREFECT1_AREA	1_MARRIED1_CHILD	1_HINCOME1_PINCOME1_JOB	1_STUDENT1_CELL	1_CELLNAME				
11309	43962.0587	1	56	10	27	0	2	1	5天街			
11392	43962.0106	1	48	0	13	0	1	3	3家次部			
12762	43963.380	1	80	10	14	0	2	0	4神奈川			
12891	43962.8619	2	92	0	11	1	1	2	1埼玉			
13620	43962.8667	1	45	0	28	0	1	4	4日藤			
18778	43962.8080	1	62	11	27	0	2	0	5天街			
18961	43962.8778	2	49	0	14	0	2	4	1神奈川			
18963	43963.3377	1	55	10	14	0	2	2	5	4神奈川		
17039	43963.3966	2	91	9	14	0	2	0	5	4神奈川		
18718	43962.9182	1	59	10	19	0	2	7	7	2	3家次部	
16067	43962.8809	1	68	11	14	0	2	1	8	8	7	4神奈川
19076	43962.8836	1	58	10	40	0	1	2	3	3	4	7日藤
20396	43963.7812	1	65	11	19	0	2	2	2	4	3家次部	
21697	43962.8443	1	46	0	12	0	1	1	10	2	11	2千原
22149	43962.9182	1	49	0	18	0	1	1	10	2	7	3家次部
22995	43963.0128	1	95	10	12	0	2	2	4	4	4	2千原
23138	43962.8472	2	47	0	12	0	1	1	1	1	12	2千原
23962	43963.3708	1	42	7	19	0	1	1	4	2	11	3家次部
23979	43963.4187	2	69	11	11	0	2	1	8	3	8	1埼玉
24434	43963.4266	2	83	9	11	0	1	1	10	10	12	1埼玉
26122	43963.8182	1	62	11	11	0	1	1	1	1	12	1埼玉
27241	43962.8626	1	91	9	19	0	1	1	4	4	3	3家次部
27393	43962.9627	1	66	11	19	0	2	2	0	2	0	3家次部
27424	43963.1169	1	67	11	11	0	2	2	10	10	11	1埼玉
27879	43962.8927	2	47	0	11	0	1	1	8	3	3	1埼玉
27884	43963.2789	1	65	11	19	0	1	1	1	1	12	3家次部
30687	43962.9686	1	68	11	14	0	2	2	0	2	12	4神奈川
32060	43962.8623	1	62	11	40	0	1	1	4	4	1	7日藤
33080	43963.1960	1	62	11	11	0	1	1	1	1	12	1埼玉
34014	43963.884	1	95	10	19	0	2	2	0	0	3	3家次部
34419	43962.9012	1	91	9	19	0	1	1	2	2	5	3家次部
38328	43963.2164	1	61	11	14	0	2	2	0	3	4	4神奈川
38370	43963.2482	2	49	0	13	0	2	2	0	0	5	3家次部

Transparent Model



Interpretable Classification

The patient is experiencing likely distress (K6), moderate depression (PHQ-9), moderate anxiety (GAD-7), high somatic symptom burden (SSS-8), and moderate loneliness (UCLA LS).

Contextualized Explanations

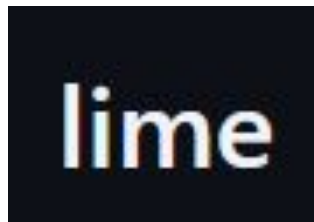
If the patient were to exercise more often, build a healthier diet, and focus more on their favorite activity, then they would experience lesser depression and anxiety.



State of the Art



SHAP



DeepLIFT



State of the Art

InterpretML

Diverse Counterfactual Explanations (DiCE) for ML

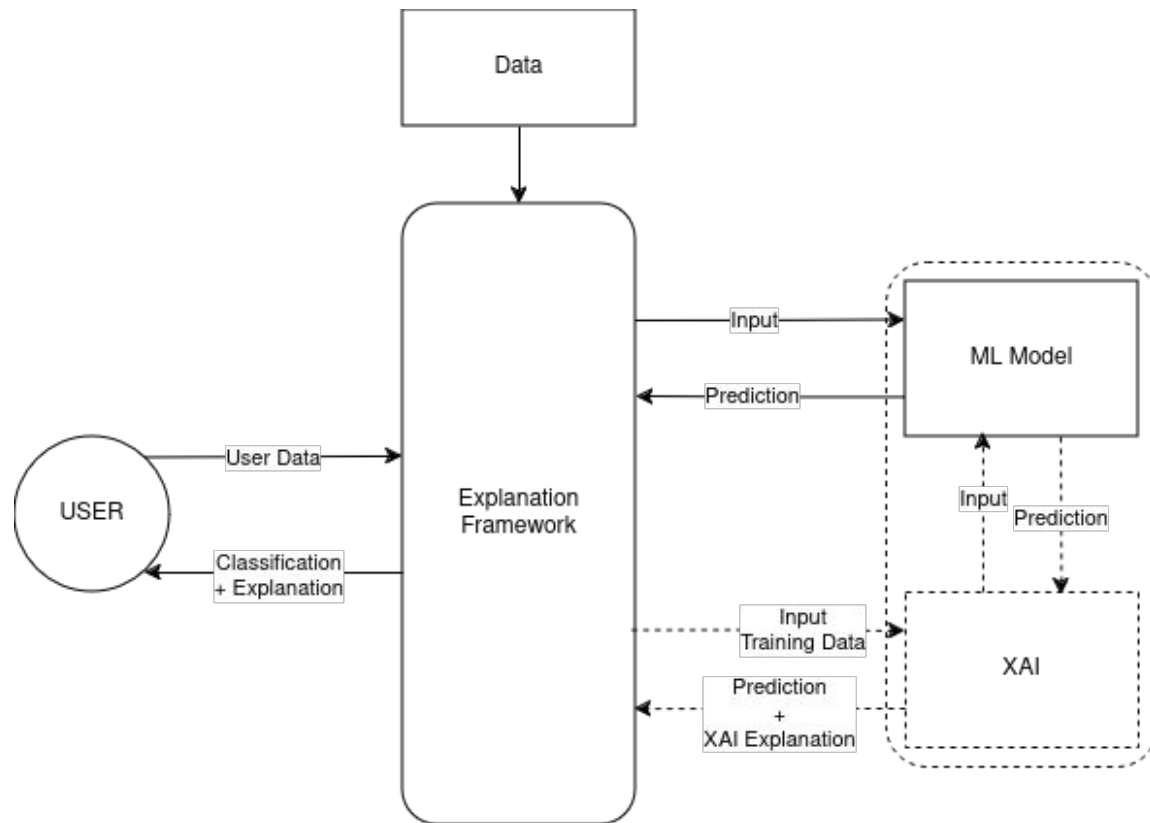
How to explain a machine learning model such that the explanation is truthful to the model and yet interpretable to people?



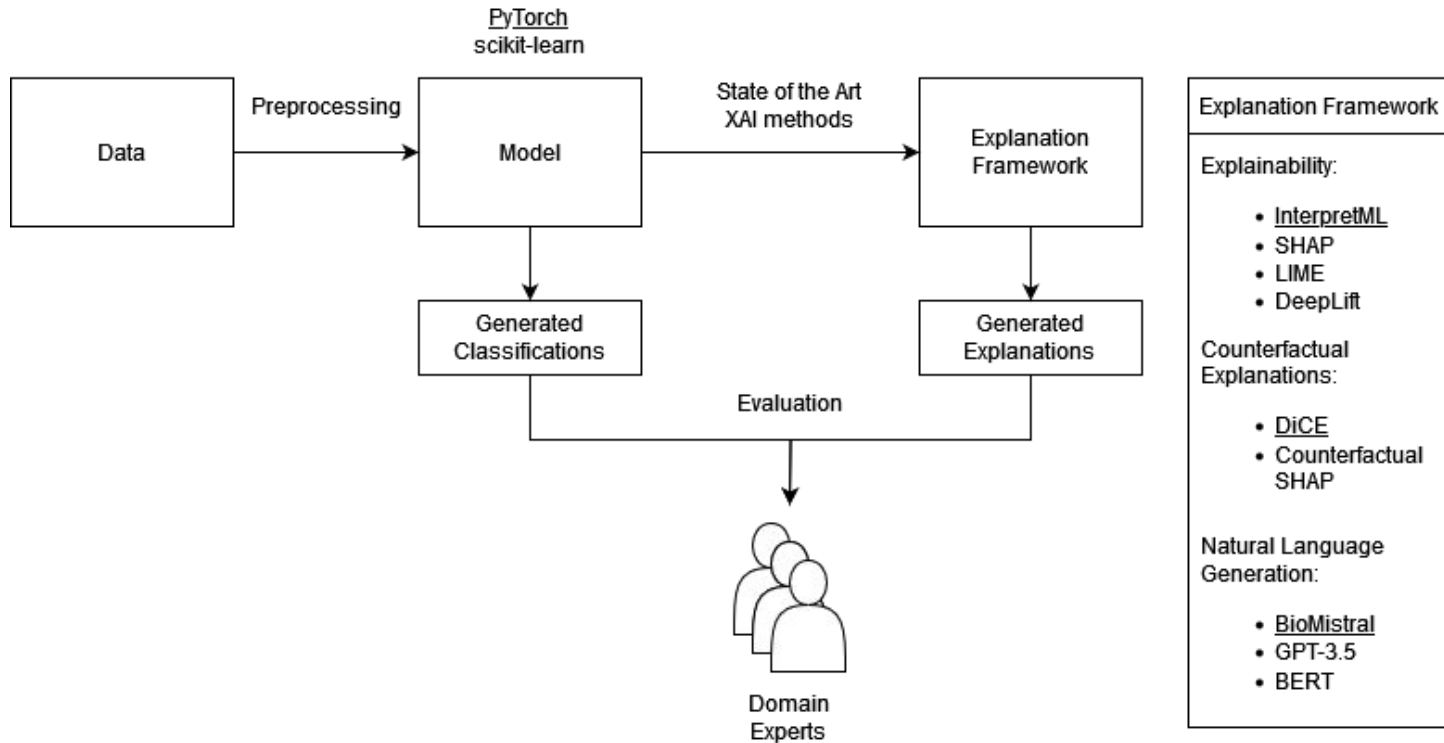
State of the Art



Theoretical Framework



Proposed Methodology





Timeline

First Semester (CS 198)	
Dates (2024)	Tasks
October 7 - October 21	Data preprocessing and visualization
October 21 - November 11	Baseline classifier
November 11 - December 2	Explanation framework

Second Semester (CS 199)	
Dates (2025)	Tasks
January 20 - February 24	Model improvement
February 24 - May 19	Web platform (visualization)



Summary

Research Problem

- The lack of transparent models used in mental health applications.

Method

- Develop an XAI using state of the art methods.

Goal

- Generate interpretable classifications and contextualized explanations for psychological profile predictions.



Summary

“... shedding light on the dynamic nature of mental health outcomes over time and underscoring the importance of continued support and interventions tailored to the changing mental health requirements of this affected population.”

(Badinlou et al., 2024)



References

Albini et al., Counterfactual Shapley Additive Explanations, 1054–1070.

Ćosić et al., AI-Based Prediction in Ex-COVID-19 Patients.

De Leon et al., "Consumption of Dietary Guidelines for Americans," 1355–1362.

Deckx, Van Den Akker, and Buntinx, "Risk Factors for Loneliness," 466–477.

Engelmann et al., "Psychological Factors Associated with Long COVID".

Generalised Anxiety Disorder Assessment (GAD-7).

Gierk et al., "Somatic Symptom Scale-8," 399–407.

Kessler et al., "Screening for Serious Mental Illness," 4–22.

Kroenke, Spitzer, and Williams, "The PHQ-9," 606–613.

Kshetry and Kantardzic, "What-If XAI Framework (WiXAI)," 169–198.

Lubben, Lubben Social Network Scale.

Lundberg and Lee, A Unified Approach to Interpreting.

Microsoft, InterpretML: A Toolkit for Understanding Machine Learning Models.

Miller, Explanation in Artificial Intelligence.

Molnar, Counterfactual Explanations.

Mothilal, Sharma, and Tan, "Diverse Counterfactual Explanations," 607–617.

Nuñez, Profiling Coping Behaviors During COVID-19.

Ribeiro, Singh, and Guestrin, "Why Should I Trust You?".

Rius Ottenheim et al., "Mental Health Deterioration from Pre- to Post-COVID".

Shrikumar, Greenside, and Kundaje, Learning Important Features.

Singh, Rani, and Srilakshmi, "Towards Explainable AI," 1–5.

Wachter, Mittelstadt, and Russell, "Counterfactual Explanations and GDPR".